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Surname

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Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced Subsidiary

Paper 2: Statistics and Mechanics

Wednesday 23 May 2018 – Morning

Time: 1 hour 15 minutes

Paper Reference

8MA0/02

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 60.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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SECTION B: MECHANICS

Unless otherwise indicated, wherever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

6. A man throws a tennis ball into the air so that, at the instant when the ball leaves his hand, the ball is 2 m above the ground and is moving vertically upwards with speed 9 m s^{-1}

The motion of the ball is modelled as that of a particle moving freely under gravity and the acceleration due to gravity is modelled as being of constant magnitude 10 m s^{-2}

The ball hits the ground T seconds after leaving the man's hand.

Using the model, find the value of T .

(4)

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Question 6 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 6 is 4 marks)



7. A train travels along a straight horizontal track between two stations, A and B .

In a model of the motion, the train starts from rest at A and moves with constant acceleration 0.3 m s^{-2} for 80 s .

The train then moves at constant velocity before it moves with a constant deceleration of 0.5 ms^{-2} , coming to rest at B .

- (a) For this model of the motion of the train between A and B ,
- (i) state the value of the constant velocity of the train,
 - (ii) state the time for which the train is decelerating,
 - (iii) sketch a velocity-time graph.

(3)

The total distance between the two stations is 4800 m.

- (b) Using the model, find the total time taken by the train to travel from A to B .

(3)

- (c) Suggest one improvement that could be made to the model of the motion of the train from A to B in order to make the model more realistic.

(1)



Question 7 continued

Handwriting practice area with 30 horizontal lines.

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 7 marks)



8. A particle, P , moves along the x -axis. At time t seconds, $t \geq 0$, the displacement, x metres, of P from the origin O , is given by $x = \frac{1}{2}t^2(t^2 - 2t + 1)$

(a) Find the times when P is instantaneously at rest.

(5)

(b) Find the total distance travelled by P in the time interval $0 \leq t \leq 2$

(3)

(c) Show that P will never move along the negative x -axis.

(2)

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Question 8 continued

Handwriting practice area with 20 horizontal lines.

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Question 8 continued

Lined area for writing the answer to Question 8.

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 10 marks)



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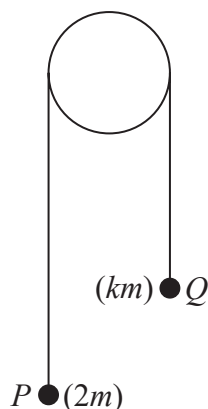


Figure 1

Two small balls, P and Q , have masses $2m$ and km respectively, where $k < 2$. The balls are attached to the ends of a string that passes over a fixed pulley. The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 1.

The system is released from rest and, in the subsequent motion, P moves downwards with an acceleration of magnitude $\frac{5g}{7}$

The balls are modelled as particles moving freely.
The string is modelled as being light and inextensible.
The pulley is modelled as being small and smooth.

Using the model,

(a) find, in terms of m and g , the tension in the string, (3)

(b) explain why the acceleration of Q also has magnitude $\frac{5g}{7}$ (1)

(c) find the value of k . (4)

(d) Identify one limitation of the model that will affect the accuracy of your answer to part (c). (1)



Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 9 marks)

TOTAL FOR SECTION B IS 30 MARKS
TOTAL FOR PAPER IS 60 MARKS

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Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Wednesday 22 May 2019

Morning

Paper Reference **8MA0-22**

**Mathematics
Advanced Subsidiary
Paper 22: Mechanics**

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 3 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. At time $t = 0$, a parachutist falls vertically from rest from a helicopter which is hovering at a height of 550 m above horizontal ground.

The parachutist, who is modelled as a particle, falls for 3 seconds before her parachute opens.

While she is falling, and before her parachute opens, she is modelled as falling freely under gravity.

The acceleration due to gravity is modelled as being 10 m s^{-2} .

- (a) Using this model, find the speed of the parachutist at the instant her parachute opens. (1)

When her parachute is open, the parachutist continues to fall vertically.

Immediately after her parachute opens, she decelerates at 12 m s^{-2} for 2 seconds before reaching a constant speed and she reaches the ground with this speed.

The total time taken by the parachutist to fall the 550 m from the helicopter to the ground is T seconds.

- (b) Sketch a speed-time graph for the motion of the parachutist for $0 \leq t \leq T$. (2)

- (c) Find, to the nearest whole number, the value of T . (5)

In a refinement of the model of the motion of the parachutist, the effect of air resistance is included before her parachute opens and this refined model is now used to find a new value of T .

- (d) How would this new value of T compare with the value found, using the initial model, in part (c)? (1)

- (e) Suggest one further refinement to the model, apart from air resistance, to make the model more realistic. (1)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 10 marks)



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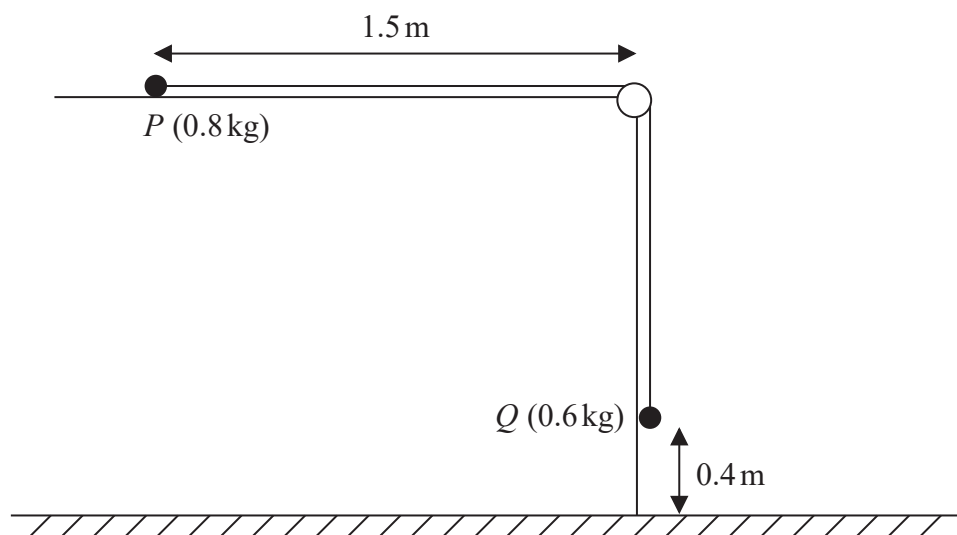


Figure 1

A small ball, P , of mass 0.8 kg , is held at rest on a smooth horizontal table and is attached to one end of a thin rope.

The rope passes over a pulley that is fixed at the edge of the table.

The other end of the rope is attached to another small ball, Q , of mass 0.6 kg , that hangs freely below the pulley.

Ball P is released from rest, with the rope taut, with P at a distance of 1.5 m from the pulley and with Q at a height of 0.4 m above the horizontal floor, as shown in Figure 1.

Ball Q descends, hits the floor and does not rebound.

The balls are modelled as particles, the rope as a light and inextensible string and the pulley as small and smooth.

Using this model,

- (a) show that the acceleration of Q , as it falls, is 4.2 m s^{-2} (5)
- (b) find the time taken by P to hit the pulley from the instant when P is released. (6)
- (c) State one limitation of the model that will affect the accuracy of your answer to part (a). (1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 12 marks)



3. A particle, P , moves along a straight line such that at time t seconds, $t \geq 0$, the velocity of P , $v \text{ m s}^{-1}$, is modelled as

$$v = 12 + 4t - t^2$$

Find

- (a) the magnitude of the acceleration of P when P is at instantaneous rest, (5)
- (b) the distance travelled by P in the interval $0 \leq t \leq 3$ (3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 8 marks)

TOTAL FOR MECHANICS IS 30 MARKS



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Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Wednesday 14 October 2020			
Afternoon		Paper Reference 8MA0/22	
Mathematics Advanced Subsidiary Paper 22: Mechanics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 3 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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1. At time $t = 0$, a small ball is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A that is 16.8 m above horizontal ground.

The speed of the ball at the instant immediately before it hits the ground for the first time is 19 m s^{-1}

The ball hits the ground for the first time at time $t = T$ seconds.

The motion of the ball, from the instant it is projected until the instant just before it hits the ground for the first time, is modelled as that of a particle moving freely under gravity.

The acceleration due to gravity is modelled as having magnitude 10 m s^{-2}

Using the model,

- (a) show that $U = 5$ (2)
- (b) find the value of T , (2)
- (c) find the time from the instant the ball is projected until the instant when the ball is 1.2 m below A . (4)
- (d) Sketch a velocity-time graph for the motion of the ball for $0 \leq t \leq T$, stating the coordinates of the start point and the end point of your graph. (2)

In a refinement of the model of the motion of the ball, the effect of air resistance on the ball is included and this refined model is now used to find the value of U .

- (e) State, with a reason, how this new value of U would compare with the value found in part (a), using the initial unrefined model. (1)
- (f) Suggest one further refinement that could be made to the model, apart from including air resistance, that would make the model more realistic. (1)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 12 marks)



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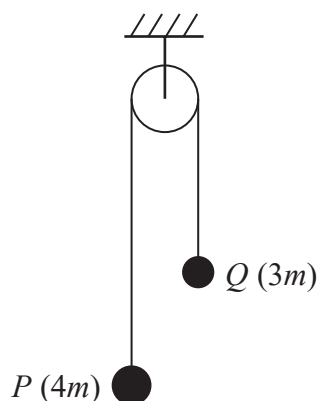


Figure 1

One end of a string is attached to a small ball P of mass $4m$.

The other end of the string is attached to another small ball Q of mass $3m$.

The string passes over a fixed pulley.

Ball P is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 1.

Ball P is released.

The string is modelled as being light and inextensible, the balls are modelled as particles, the pulley is modelled as being smooth and air resistance is ignored.

- (a) Using the model, find, in terms of m and g , the magnitude of the force exerted on the pulley by the string while P is falling and before Q hits the pulley. (8)
- (b) State one limitation of the model, apart from ignoring air resistance, that will affect the accuracy of your answer to part (a). (1)

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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 9 marks)



3. A particle P moves along a straight line such that at time t seconds, $t \geq 0$, after leaving the point O on the line, the velocity, $v \text{ m s}^{-1}$, of P is modelled as

$$v = (7 - 2t)(t + 2)$$

- (a) Find the value of t at the instant when P stops accelerating. (4)

- (b) Find the distance of P from O at the instant when P changes its direction of motion. (5)

In this question, solutions relying on calculator technology are not acceptable.



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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(Total for Question 3 is 9 marks)

TOTAL FOR MECHANICS IS 30 MARKS



1. At time $t = 0$, a small stone is thrown vertically upwards with speed 14.7 m s^{-1} from a point A .

At time $t = T$ seconds, the stone passes through A , moving downwards.

The stone is modelled as a particle moving freely under gravity throughout its motion.

Using the model,

- (a) find the value of T , (2)
- (b) find the total distance travelled by the stone in the first 4 seconds of its motion. (4)
- (c) State one refinement that could be made to the model, apart from air resistance, that would make the model more realistic. (1)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. A particle P moves along a straight line.

At time t seconds, the velocity $v \text{ m s}^{-1}$ of P is modelled as

$$v = 10t - t^2 - k \quad t \geq 0$$

where k is a constant.

- (a) Find the acceleration of P at time t seconds.

(2)

The particle P is instantaneously at rest when $t = 6$

- (b) Find the other value of t when P is instantaneously at rest.

(4)

- (c) Find the total distance travelled by P in the interval $0 \leq t \leq 6$

(4)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

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(Total for Question 2 is 10 marks)



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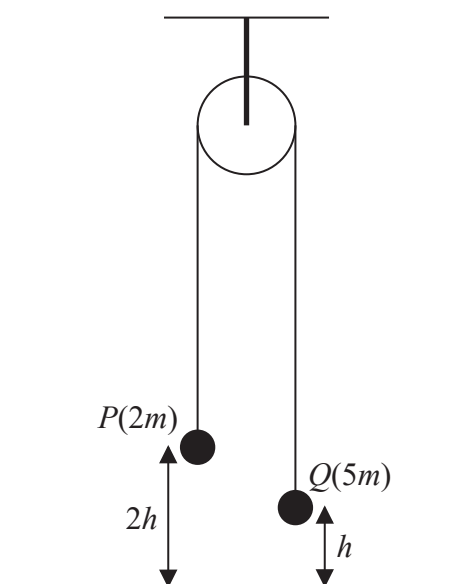


Figure 1

A ball P of mass $2m$ is attached to one end of a string.

The other end of the string is attached to a ball Q of mass $5m$.

The string passes over a fixed pulley.

The system is held at rest with the balls hanging freely and the string taut.

The hanging parts of the string are vertical with P at a height $2h$ above horizontal ground and with Q at a height h above the ground, as shown in Figure 1.

The system is released from rest.

In the subsequent motion, Q does not rebound when it hits the ground and P does not hit the pulley.

The balls are modelled as particles.

The string is modelled as being light and inextensible.

The pulley is modelled as being small and smooth.

Air resistance is modelled as being negligible.

Using this model,

- (a) (i) write down an equation of motion for P ,
 (ii) write down an equation of motion for Q ,
 (4)
- (b) find, in terms of h only, the height above the ground at which P first comes to instantaneous rest.
 (7)
- (c) State one limitation of modelling the balls as particles that could affect your answer to part (b).
 (1)

In reality, the string will not be inextensible.

- (d) State how this would affect the accelerations of the particles.
 (1)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

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(Total for Question 3 is 13 marks)

TOTAL FOR MECHANICS IS 30 MARKS



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Candidate Number

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Pearson Edexcel Level 3 GCE

Paper
reference

8MA0/22

Mathematics

Advanced Subsidiary

PAPER 22: Mechanics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Instructions

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Answers without working may not gain full credit.
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

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- (d) Explain how this would affect your answer to part (a). (1)

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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2. A train travels along a straight horizontal track from station P to station Q .

In a model of the motion of the train, at time $t = 0$ the train starts from rest at P , and moves with constant acceleration until it reaches its maximum speed of 25 m s^{-1}

The train then travels at this constant speed of 25 m s^{-1} before finally moving with constant deceleration until it comes to rest at Q .

The time spent decelerating is four times the time spent accelerating.

The journey from P to Q takes 700 s.

Using the model,

- (a) sketch a speed-time graph for the motion of the train between the two stations P and Q . (1)

The distance between the two stations is 15 km.

Using the model,

- (b) show that the time spent accelerating by the train is 40 s, (3)

- (c) find the acceleration, in m s^{-2} , of the train, (1)

- (d) find the speed of the train 572 s after leaving P . (2)

- (e) State one limitation of the model which could affect your answers to parts (b) and (c). (1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. A fixed point O lies on a straight line.

A particle P moves along the straight line.

At time t seconds, $t \geq 0$, the distance, s metres, of P from O is given by

$$s = \frac{1}{3}t^3 - \frac{5}{2}t^2 + 6t$$

- (a) Find the acceleration of P at each of the times when P is at instantaneous rest. (6)
- (b) Find the total distance travelled by P in the interval $0 \leq t \leq 4$ (3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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(Total for Question 4 is 6 marks)

TOTAL FOR MECHANICS IS 30 MARKS



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Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Thursday 25 May 2023

Afternoon

Paper
reference

8MA0/22

Mathematics

**Advanced Subsidiary
PAPER 22: Mechanics**

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Instructions

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Information

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- The total mark for this part of the examination is 30. There are 4 questions.
- The marks for **each** question are shown in brackets
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Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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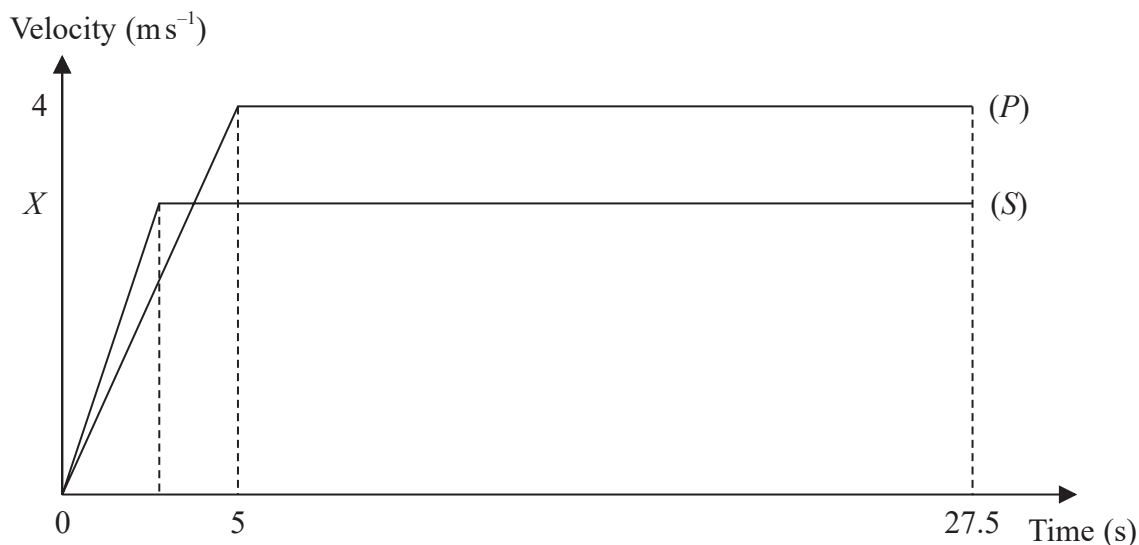


Figure 1

Two children, Pat (*P*) and Sam (*S*), run a race along a straight horizontal track.

Both children start from rest at the same time and cross the finish line at the same time.

In a model of the motion:

Pat accelerates at a constant rate from rest for 5 s until reaching a speed of 4 m s^{-1} and then maintains a constant speed of 4 m s^{-1} until crossing the finish line.

Sam accelerates at a constant rate of 1 m s^{-2} from rest until reaching a speed of $X \text{ m s}^{-1}$ and then maintains a constant speed of $X \text{ m s}^{-1}$ until crossing the finish line.

Both children take 27.5 s to complete the race.

The velocity-time graphs shown in Figure 1 describe the model of the motion of each child from the instant they start to the instant they cross the finish line together.

Using the model,

- explain why the areas under the two graphs are equal, (1)
- find the acceleration of Pat during the first 5 seconds, (1)
- find, in metres, the length of the race, (2)
- find the value of X , giving your answer to 3 significant figures. (4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. A small stone is projected vertically upwards with speed 39.2 m s^{-1} from a point O .

The stone is modelled as a particle moving freely under gravity from when it is projected until it hits the ground 10 s later.

Using the model, find

- (a) the height of O above the ground, (3)
- (b) the total length of time for which the speed of the stone is less than or equal to 24.5 m s^{-1} (3)
- (c) State one refinement that could be made to the model that would make your answer to part (a) more accurate. (1)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 7 marks)



3.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

A fixed point O lies on a straight line.

A particle P moves along the straight line such that at time t seconds, $t \geq 0$, after passing through O , the velocity of P , $v \text{ m s}^{-1}$, is modelled as

$$v = 15 - t^2 - 2t$$

- (a) Verify that P comes to instantaneous rest when $t = 3$ (1)
- (b) Find the magnitude of the acceleration of P when $t = 3$ (3)
- (c) Find the total distance travelled by P in the interval $0 \leq t \leq 4$ (4)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 8 marks)

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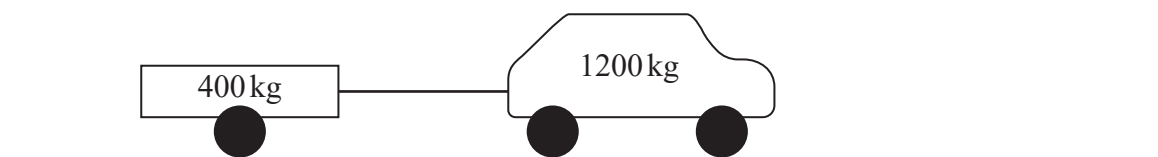


Figure 2

A car of mass 1200 kg is towing a trailer of mass 400 kg along a straight horizontal road using a tow rope, as shown in Figure 2.

The rope is horizontal and parallel to the direction of motion of the car.

- The resistance to motion of the car is modelled as a constant force of magnitude $2R$ newtons
- The resistance to motion of the trailer is modelled as a constant force of magnitude R newtons
- The rope is modelled as being light and inextensible
- The acceleration of the car is modelled as $a \text{ m s}^{-2}$

The driving force of the engine of the car is 7400 N and the tension in the tow rope is 2400 N.

Using the model,

(a) find the value of a

(5)

In a refined model, the rope is modelled as having mass and the acceleration of the car is found to be $a_1 \text{ m s}^{-2}$

(b) State how the value of a_1 compares with the value of a

(1)

(c) State one limitation of the model used for the resistance to motion of the car.

(1)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 4 is 7 marks)

TOTAL FOR MECHANICS IS 30 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	
Pearson Edexcel Level 3 GCE			
Thursday 23 May 2024			
Afternoon		Paper reference	8MA0/22
Mathematics Advanced Subsidiary PAPER 22: Mechanics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

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Advice

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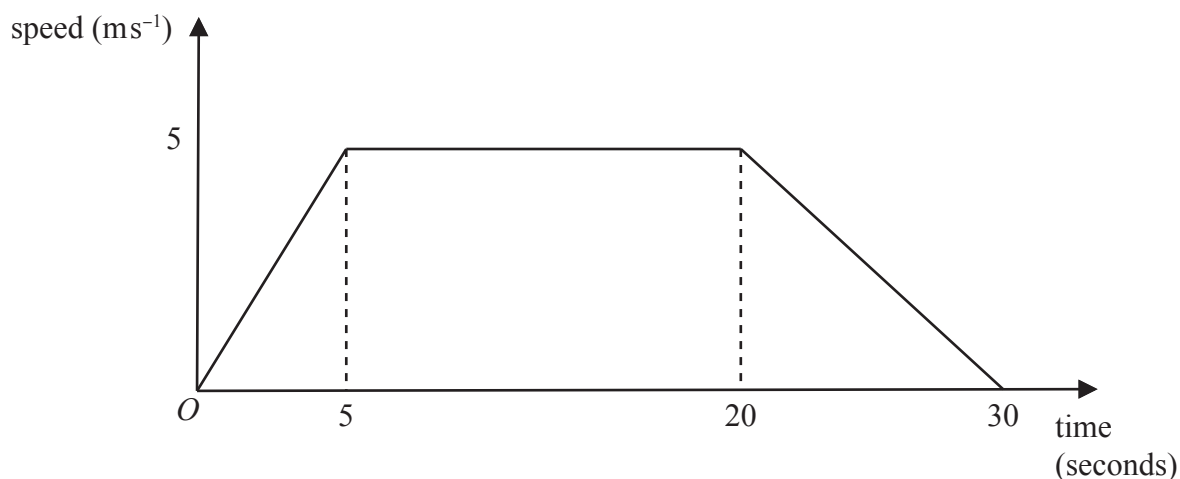
**Figure 1**

Figure 1 shows the speed-time graph for the journey of a car moving in a long queue of traffic on a straight horizontal road.

At time $t = 0$, the car is at rest at the point A .

The car then accelerates uniformly for 5 seconds until it reaches a speed of 5 ms^{-1}

For the next 15 seconds the car travels at a constant speed of 5 ms^{-1}

The car then decelerates uniformly until it comes to rest at the point B .

The total journey time is 30 seconds.

(a) Find the distance AB . (3)

(b) Sketch a distance-time graph for the journey of the car from A to B . (3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

A particle is moving along a straight line.

At time t seconds, $t > 0$, the velocity of the particle is $v \text{ ms}^{-1}$, where

$$v = 2t - 7\sqrt{t} + 6$$

- (a) Find the acceleration of the particle when $t = 4$ (3)

When $t = 1$ the particle is at the point X .

When $t = 2$ the particle is at the point Y .

Given that the particle does not come to instantaneous rest in the interval $1 < t < 2$

- (b) show that $XY = \frac{1}{3}(41 - 28\sqrt{2})$ metres. (4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A particle P is moving on a smooth horizontal surface under the action of two forces.

Given that

- the mass of P is 2 kg
- the two forces are $(2\mathbf{i} + 4\mathbf{j})\text{ N}$ and $(c\mathbf{i} - 2\mathbf{j})\text{ N}$, where c is a constant
- the magnitude of the acceleration of P is $\sqrt{5}\text{ ms}^{-2}$

find the two possible values of c .

(5)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



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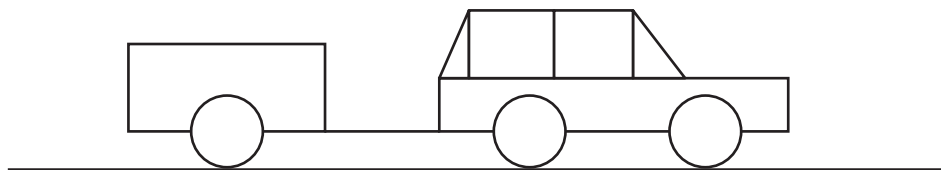


Figure 2

Figure 2 shows a car towing a trailer along a straight horizontal road.

The mass of the car is 800 kg and the mass of the trailer is 600 kg.

The trailer is attached to the car by a towbar which is parallel to the road and parallel to the direction of motion of the car and the trailer.

The towbar is modelled as a light rod.

The resistance to the motion of the car is modelled as a constant force of magnitude 400 N.

The resistance to the motion of the trailer is modelled as a constant force of magnitude R newtons.

The engine of the car is producing a constant driving force that is horizontal and of magnitude 1740 N.

The acceleration of the car is 0.6 ms^{-2} and the tension in the towbar is T newtons.

Using the model,

(a) show that $R = 500$ (3)

(b) find the value of T . (3)

At the instant when the speed of the car and the trailer is 12.5 ms^{-1} , the towbar breaks.

The trailer moves a further distance d metres before coming to rest.

The resistance to the motion of the trailer is modelled as a constant force of magnitude 500 N.

Using the model,

(c) show that, after the towbar breaks, the deceleration of the trailer is $\frac{5}{6} \text{ ms}^{-2}$ (1)

(d) find the value of d . (3)

In reality, the distance d metres is likely to be different from the answer found in part (d).

(e) Give two **different** reasons why this is the case. (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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(Total for Question 4 is 12 marks)

TOTAL FOR MECHANICS IS 30 MARKS



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Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Friday 23 May 2025

Afternoon	Paper reference	8MA0/22
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Mathematics

Advanced Subsidiary

PAPER 22: Mechanics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

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- The total mark for this part of the examination is 30. There are 4 questions.
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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

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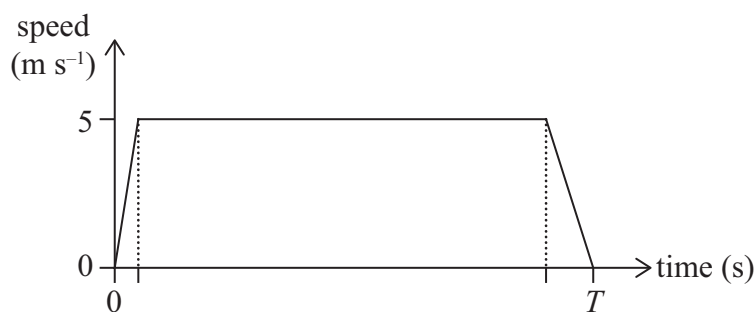


Figure 1

Figure 1 shows a sketch of the speed-time graph for a model of the motion of a runner travelling along a straight horizontal road from a point A to a point B .

The distance from A to B is 400 metres.

In the model of the motion, the runner

- starts from rest at A at time $t = 0$
- then moves with constant acceleration for 5 seconds, reaching a maximum speed of 5 m s^{-1}
- then travels at a constant speed of 5 m s^{-1}
- then moves with constant deceleration for 15 seconds, until coming to rest at B
- travels from A to B in T seconds

(a) Find the value of T .

(3)

(b) State **one** reason why the **actual** time taken to travel from A to B might not be T seconds.

(1)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 4 marks)



2. A particle P has mass 0.25 kg .

The particle moves on a smooth horizontal surface under the action of two horizontal forces $\mathbf{F}_1$ and $\mathbf{F}_2$

Given that $\mathbf{F}_1 = (3\mathbf{i} - 5\mathbf{j})\text{ N}$ and $\mathbf{F}_2 = (-6\mathbf{i} + 9\mathbf{j})\text{ N}$

- (a) find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the **resultant force** acting on P , (1)

- (b) find the **magnitude** of the acceleration of P . (3)

A third horizontal force, $\mathbf{F}_3$, is now applied to P .

Under the action of $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, the particle moves with **constant velocity**.

- (c) Find $\mathbf{F}_3$, giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$. (1)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. A package P of mass 2 kg is held at rest on a horizontal table.
 One end of a thin rope is attached to P .
 The rope passes over a pulley that is fixed at the edge of the table.
 The other end of the rope is attached to a package Q of mass 3 kg that hangs freely below the pulley.

Package P is 1.2 m from the pulley.

Package Q is 0.4 m above a horizontal floor and the rope is taut, as shown in Figure 2 on page 7.

Package P is released from rest and Q moves downwards.

In an initial model of the motion of P and Q

- the table is modelled as being smooth
- the packages are modelled as particles
- air resistance is modelled as being negligible
- the pulley is modelled as being small and smooth
- the rope is modelled as being light and inextensible

Using this model,

- (a) write down an equation of motion for Q as it moves downwards, (2)
- (b) find the acceleration of Q as it moves downwards, (3)
- (c) find the speed of Q at the instant when Q hits the floor. (2)

Given that K seconds **after** Q hits the floor, P hits the pulley

- (d) use the model to find the value of K . (2)

In a refinement of the model, there is a **resistance** which acts against the motion of P .

This refined model is now used to find the acceleration of Q as it moves downwards.

- (e) State, **with a reason**, whether this **new** acceleration will be greater than, equal to, or less than, the acceleration found in part (b). (1)
- (f) Suggest one **further** refinement that could be made to the model, apart from including air resistance, that would make the model more realistic. (1)



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Question 3 continued

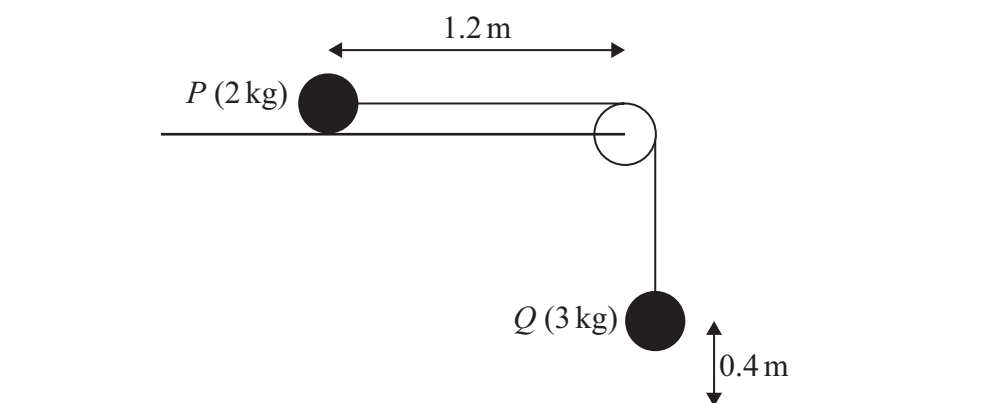


Figure 2

Lined area for student response.

Question 3 continued

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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 11 marks)



4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

A fixed point O lies on a straight line.

A particle P moves along the straight line.

At time t seconds, $t \geq 0$, the distance, s metres, of P from O is given by

$$s = 2t^3 - 12t^2 + 18t$$

- (a) Find the **values** of t for which P is instantaneously at rest. (4)
- (b) Find the **total** distance travelled by P in the interval $0 \leq t \leq 4$ (3)
- (c) Find the maximum speed of P in the interval $1 \leq t \leq 3$ (3)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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(Total for Question 4 is 10 marks)

TOTAL FOR MECHANICS IS 30 MARKS



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Level 3 GCE**

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Mock Paper Set 1

Paper Reference **8MA0/22**

Mathematics

**Advanced Subsidiary
Paper 22: Mechanics**

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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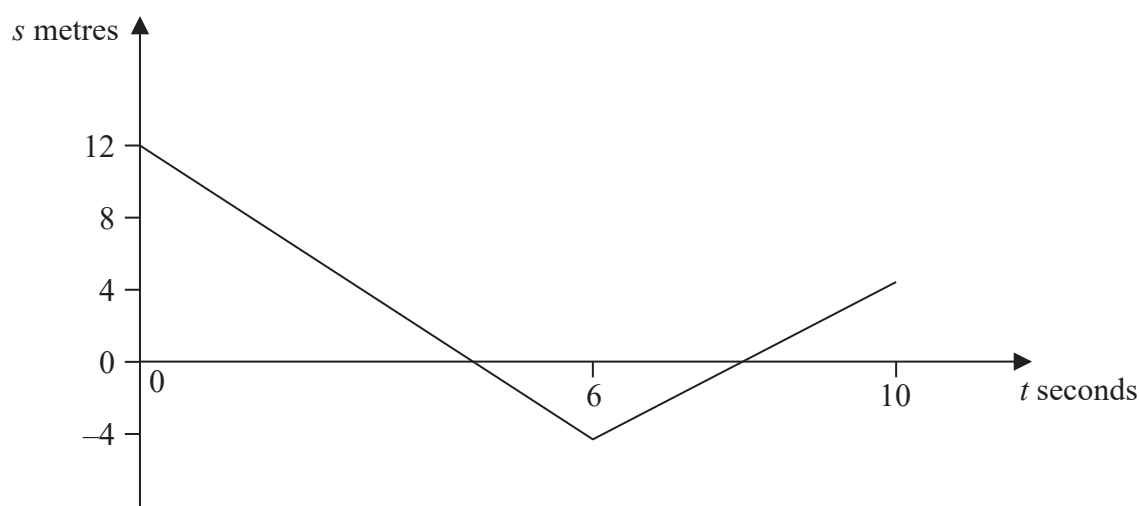


Figure 1

A toy car moves along a straight line.

The line contains the fixed point O

The toy car is modelled as a particle.

Figure 1 shows the **displacement-time** graph for a model of the motion of the toy car, where s metres is the displacement of the toy car from O at time t seconds.

Using the model,

- (a) find the speed of the toy car in the interval $0 \leq t \leq 6$ (1)
- (b) find the average speed of the toy car in the interval $0 \leq t \leq 10$ (2)
- (c) sketch the **velocity-time** graph for the motion of the toy car in the interval $0 \leq t \leq 10$ (2)
- (d) (i) Referring to your sketch, or otherwise, describe what happens to the motion of the toy car at the instant when $t = 6$
- (ii) Explain why your answer to (i) represents a limitation of the model of the motion of the toy car at $t = 6$ (2)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 7 marks)



2.

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 5 kg is moving on a smooth horizontal plane.

The particle P is moving under the action of two forces $(\mathbf{i} + 2\mathbf{j})\text{ N}$ and $(-c\mathbf{i} - 10\mathbf{j})\text{ N}$, where c is a positive constant.

The magnitude of the acceleration of P is 2 m s^{-2}

Find the value of c

(5)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. A fixed point O lies on a straight line.

A particle P moves along this line.

At time $t = 0$, particle P passes through O

At time t seconds, $t \geq 0$, the velocity, $v \text{ m s}^{-1}$, of P is

$$v = 6 + 4t^{\frac{3}{2}} - 3t^2$$

- (a) Find the acceleration of P at time t seconds.

(2)

- (b) Find the distance of P from O at the instant when $t > 0$ and the acceleration of P is zero.

(6)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 8 marks)



4.

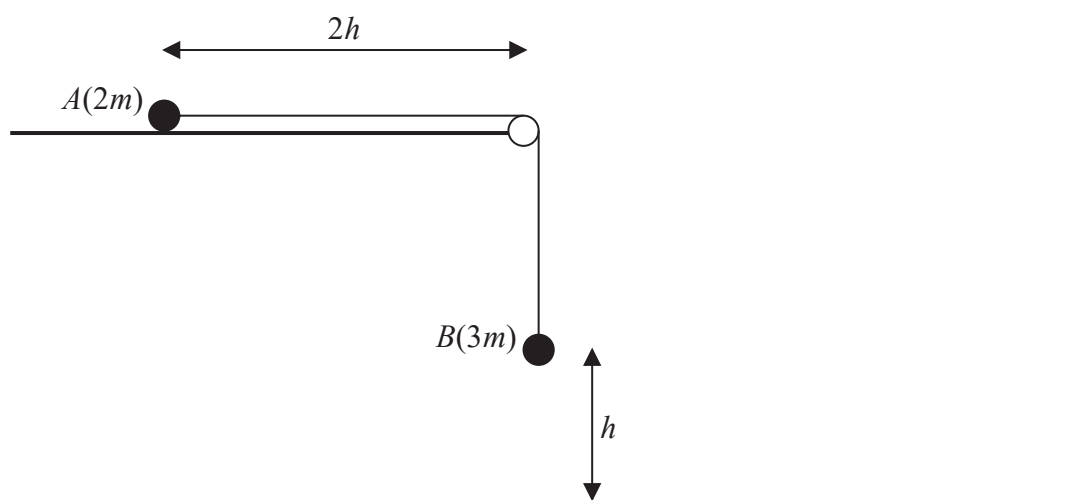


Figure 2

A small smooth light pulley is fixed to the edge of a rough horizontal table.

Two particles A and B are connected by a string which passes over the pulley.

- Particle A has mass $2m$
- Particle B has mass $3m$
- Particle A is held at rest on the rough table at a distance of $2h$ from the pulley
- Particle B hangs freely below the pulley, with the string taut, at a height h above a horizontal floor, as shown in Figure 2

Particle A is released from rest and begins to move along the table.

In an initial model,

- the string is modelled as being light and inextensible
- the resistance to motion that the table exerts on A is modelled as a constant force of magnitude $\frac{1}{2}mg$
- air resistance is modelled as being negligible

This model predicts that in the subsequent motion, B first strikes the floor with speed V

Using this initial model,

- write down an equation of motion for A (2)
- find V in terms of g and h (6)

Question 4 continues on page 11



Question 4 continued

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(Total for Question 4 is 10 marks)

TOTAL FOR MECHANICS IS 30 MARKS



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Pearson Edexcel Level 3 GCE

Paper
reference

8MA0/22

Mathematics

Advanced Subsidiary

PAPER 22: Mechanics

Mock Set 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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1. A car is moving along a straight road.

- At time $t = 0$, the speed of the car is 18 m s^{-1}
- The car decelerates at a constant rate for 5 seconds from time $t = 0$, reducing its speed from 18 m s^{-1} to $V \text{ m s}^{-1}$
- The car then continues at a constant speed of $V \text{ m s}^{-1}$ for 15 seconds
- The car then accelerates at a constant rate for 5 seconds, increasing its speed from $V \text{ m s}^{-1}$ to 16 m s^{-1}

(a) Sketch a speed-time graph for the motion of the car from $t = 0$ to $t = 25$ seconds. (2)

Between $t = 0$ to $t = 25$ seconds, the car travels a total distance of 325 m.

(b) Find the value of V . (4)

(c) Find the deceleration of the car. (2)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. A particle P moves along the positive x -axis.

At time t seconds, $t \geq 0$

- the distance of P from the origin O is x metres
- the velocity of P is $v \text{ ms}^{-1}$ in the positive x direction
- the acceleration of P is $a \text{ ms}^{-2}$

Given that $x = 4t^3 - 21t^2 + 36t + 1$

- (a) find v in terms of t , (2)
- (b) find the distance of P from O at the instant when P is at rest for the second time, (4)
- (c) find a in terms of t . (2)



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 8 marks)



3. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass m kg is moving on a smooth horizontal plane.

The particle P is moving under the action of two forces $(2\mathbf{i} - \mathbf{j})\text{N}$ and $(c\mathbf{i} - 13.4\mathbf{j})\text{N}$, where c is a constant.

The acceleration of P is $(10\mathbf{i} - 24\mathbf{j})\text{ms}^{-2}$

Find the value of c .

(4)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 4 marks)



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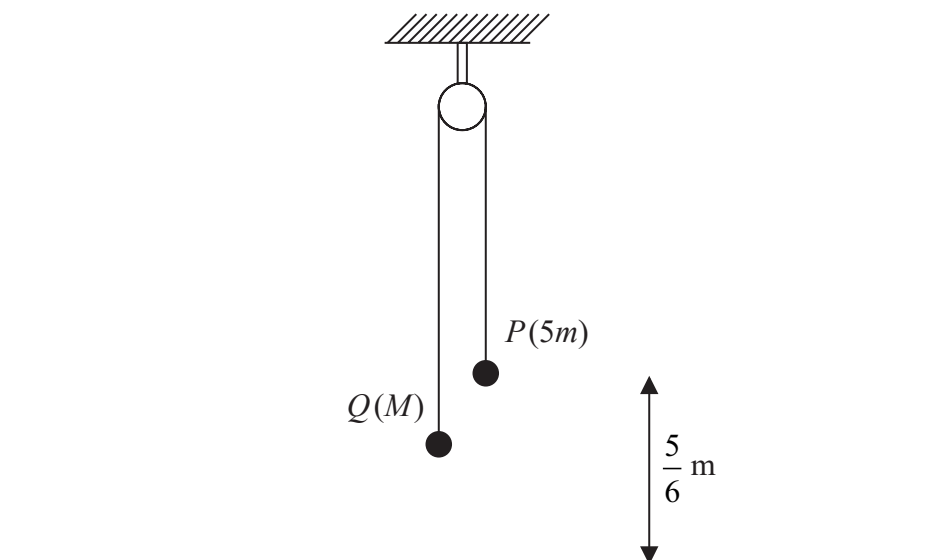


Figure 1

A particle P has mass $5m$.

A particle Q has mass M , where $M < 5m$.

The particles P and Q are connected by a light inextensible string.

The string passes over a small smooth light pulley that is fixed above horizontal ground.

Particle P is held at rest at a distance $\frac{5}{6}$ m above the ground.

Particle Q hangs freely with the string taut, as shown in Figure 1.

At time $t = 0$, P is released from rest and begins to move downwards.

The particle P reaches the ground before Q reaches the pulley.

As P moves downwards, the tension in the string is $\frac{10mg}{3}$

In an initial model,

- air resistance is negligible
- P hits the ground at time $t = T$ seconds

Using the model,

(a) find M in terms of m (6)

(b) find the value of T (2)

In a refinement to the model,

- air resistance is included
- P hits the ground at time $t = T_1$ seconds

(c) State which is greater, T or T_1 . You must give a reason for your answer. (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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(Total for Question 4 is 10 marks)

TOTAL FOR PAPER IS 30 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Mock Paper Set 3

Paper reference	8MA0/22
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Mathematics

Advanced Subsidiary

PAPER 22: Mechanics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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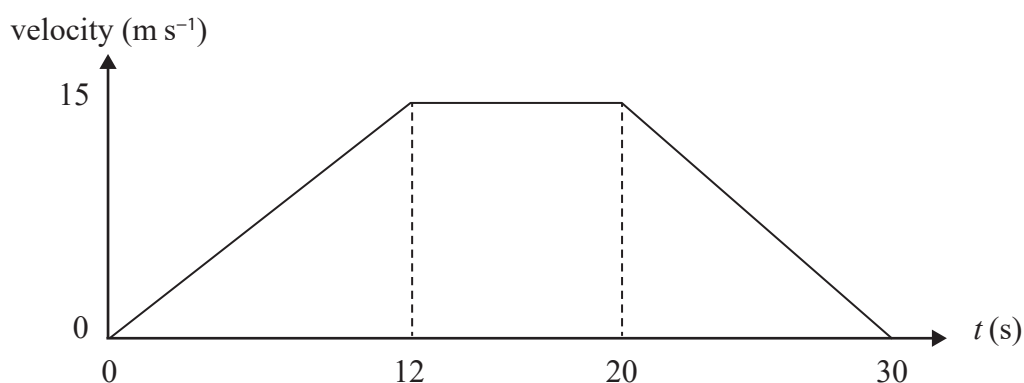


Figure 1

Figure 1 shows a velocity-time graph for the motion of a car.

The car is moving along a straight horizontal road between two points A and B ,

- at time $t = 0$, the car starts from rest at A
- the car accelerates at a constant rate for 12 seconds until it reaches a velocity of 15 m s^{-1}
- the car then maintains a constant velocity of 15 m s^{-1} for T seconds
- the car then decelerates at a constant rate
- the car comes to rest at B

(a) Write down the value of T . (1)

(b) Find the acceleration of the car between $t = 0$ and $t = 12$ (2)

(c) Find the distance AB . (3)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2.

[In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P is on a smooth horizontal plane.

The particle P is at rest under the action of three forces:

$$\mathbf{F}_1 = (-\mathbf{i} + 2\mathbf{j}) \text{ N},$$

$$\mathbf{F}_2 = (3\mathbf{i} - 4\mathbf{j}) \text{ N},$$

$$\mathbf{F}_3 = (c\mathbf{i} + 2\mathbf{j}) \text{ N}, \text{ where } c \text{ is a constant.}$$

Given that P is in equilibrium,

(a) find the value of c .

(2)

The force $\mathbf{F}_3$ is now removed.

Given that the mass of P is 0.5 kg,

(b) find the acceleration of P , giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

A particle P moves along the x -axis.

At time t seconds, $t \geq 0$

- the distance of P from the origin O is x metres
- the velocity of P is $v \text{ ms}^{-1}$

When $t = 0$, P passes through O .

Given that

$$v = 6t^2 + 4t + 1$$

- (a) find the acceleration of P in terms of t , where $t \geq 0$

(2)

- (b) find the distance of P from O when $t = 3$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



- Using the model,

- (b) find the total time from when the stone leaves A to when the stone first hits the ground.
- (3)**

- to include air resistance
- to use a more accurate value for g

(c) State **one other** possible refinement of the model that would make it more realistic. **(1)**

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



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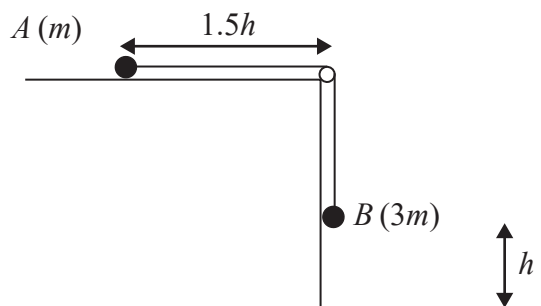


Figure 2

Two particles, A and B , are connected by a string.

Particle A has mass m .

Particle B has mass $3m$.

The string passes over a small smooth light pulley.

The pulley is fixed at the edge of a smooth horizontal table.

Particle A is held at rest on the table at a distance of $1.5h$ from the pulley.

Particle B hangs freely with the string taut, at a height h above the horizontal floor, as shown in Figure 2.

Particle A is released from rest and begins to move along the table.

In an initial model,

- the string is light and inextensible
- air resistance is ignored
- B hits the floor with speed V

Using the model,

(a) show that the tension in the string is $\frac{3mg}{4}$ (5)

(b) find V in terms of g and h (2)

One limitation of the model is that it ignores air resistance.

(c) State **one other** limitation of the model. (1)

In a refinement to the model,

- air resistance is **not** ignored
- B hits the floor with speed W

(d) State which is greater, V or W , giving a reason for your answer. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 9 marks)

TOTAL FOR MECHANICS IS 30 MARKS



Pearson Edexcel Level 3 GCE

Mathematics

Advanced Subsidiary

Paper 2: Statistics and Mechanics

Sample assessment material for first teaching
September 2017

Time: 1 hour 15 minutes

Paper Reference(s)

8MA0/02

You must have:

Mathematical Formulae and Statistical Tables
Calculator

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
- *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 9 questions in this question paper. The total mark for this paper is 60.
- The marks for each question are shown in brackets
- *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

SECTION B: MECHANICS

Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

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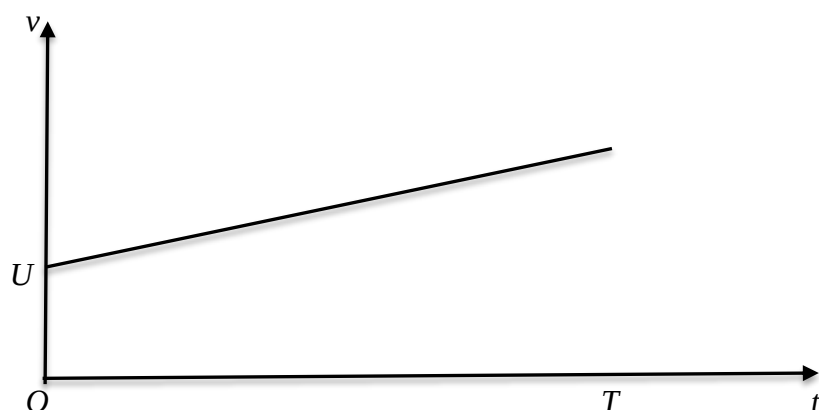


Figure 1

A car moves along a straight horizontal road. At time $t = 0$, the velocity of the car is $U \text{ m s}^{-1}$. The car then accelerates with constant acceleration $a \text{ m s}^{-2}$ for T seconds. The car travels a distance D metres during these T seconds.

Figure 1 shows the velocity-time graph for the motion of the car for $0 \leq t \leq T$.

Using the graph, show that $D = UT + \frac{1}{2}aT^2$.

(No credit will be given for answers which use any of the kinematics (*suvat*) formulae listed under Mechanics in the AS Mathematics section of the formulae booklet.)

(4)

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7. A car is moving along a straight horizontal road with constant acceleration.

There are three points A , B and C , in that order, on the road, where $AB = 22$ m and $BC = 104$ m.

The car takes 2 s to travel from A to B and 4 s to travel from B to C .

Find

- (i) the acceleration of the car,
- (ii) the speed of the car at the instant it passes A .

(7)

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Question 7 Continued

(Total for Question 7 is 7 marks)

8. A bird leaves its nest at time $t = 0$ for a short flight along a straight line.

The bird then returns to its nest.

The bird is modelled as a particle moving in a straight horizontal line.

The distance, s metres, of the bird from its nest at time t seconds is given by

$$s = \frac{1}{10}(t^4 - 20t^3 + 100t^2), \text{ where } 0 \leq t \leq 10$$

(a) Explain the restriction, $0 \leq t \leq 10$

(3)

(b) Find the distance of the bird from the nest when the bird first comes to instantaneous rest.

(6)

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(Total for Question 3 is 5 marks)

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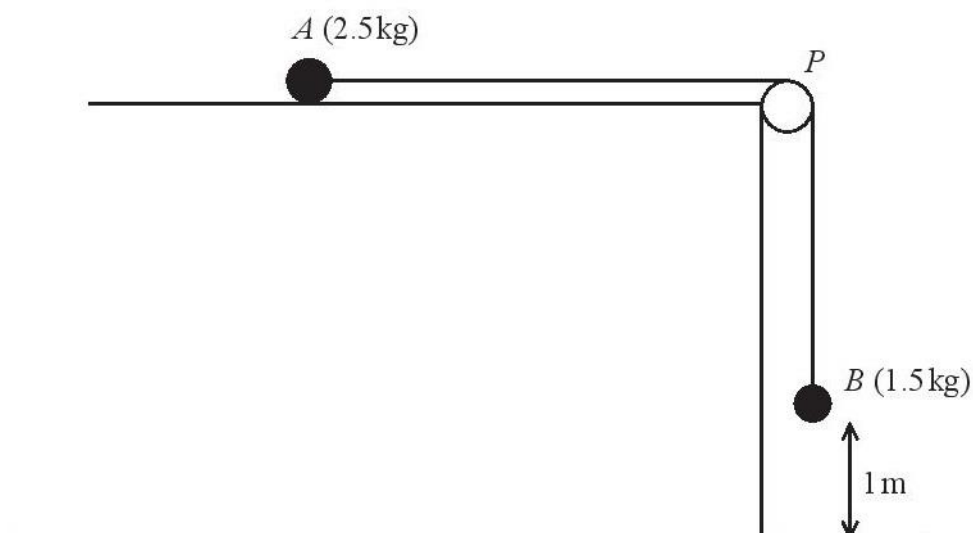


Figure 2

A small ball A of mass 2.5 kg is held at rest on a rough horizontal table.

The ball is attached to one end of a string.

The string passes over a pulley P which is fixed at the edge of the table. The other end of the string is attached to a small ball B of mass 1.5 kg hanging freely, vertically below P and with B at a height of 1 m above the horizontal floor.

The system is released from rest, with the string taut, as shown in Figure 2.

The resistance to the motion of A from the rough table is modelled as having constant magnitude 12.7 N . Ball B reaches the floor before ball A reaches the pulley.

The balls are modelled as particles, the string is modelled as being light and inextensible and the pulley is modelled as being small and smooth.

(a) (i) Write down an equation of motion for A .

(ii) Write down an equation of motion for B .

(4)

(b) Hence find the acceleration of B .

(2)

(c) Using the model, find the time it takes, from release, for B to reach the floor.

(2)

It was found that it actually took 2.3 seconds for ball B to reach the floor.

(d) Using this information

(i) comment on the appropriateness of using the model to find the time it takes ball B to reach the floor, justifying your answer.

(ii) suggest one improvement that could be made in the model.

(2)

[illegible]

[illegible]

[illegible]

TOTAL FOR PAPER IS 60 MARKS

Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced Subsidiary

Paper 2: Statistics and Mechanics

Specimen Paper

Time: 1 hour 15 minutes

Paper Reference

8MA0/02

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 60.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

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SECTION B: MECHANICS

Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

5. A small ball is projected vertically upwards from a point A which is 19.6 m above the ground. The ball strikes the ground, for the first time, 4 s later.

The motion of the ball is modelled as that of a particle moving freely under gravity.

(a) Use the model to find the speed of the ball as it hits the ground for the first time.

(3)

The ball rebounds from the ground with a vertical speed of 14.7 m s^{-1} and next comes to instantaneous rest at the point B .

(b) Use the model to find the height of B above the ground.

(2)

In a refined model of the motion of the ball, the effect of air resistance is included and this refined model is now used to find the speed of the ball as it hits the ground for the first time

(c) How would this new value of the speed of the ball as it hits the ground for the first time compare with the value found using the initial model in part (a)?

(1)

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6. A car travels along a straight horizontal road between two sets of traffic lights.
The distance between the two sets of traffic lights is 1500 m.

In a model of the journey, the car leaves the first set of traffic lights, accelerating uniformly from rest until it reaches a speed of $V \text{ m s}^{-1}$, then immediately decelerates uniformly until it comes to rest at the second set of traffic lights.

The car completes the journey between the two sets of lights in 120 s.

- (a) Sketch, on the diagram below, a velocity-time graph which represents the above model of the journey of the car between the two sets of traffic lights. (2)

- (b) Using the model, find the value of V . (2)

It is given that the car accelerates uniformly for T seconds.

- (c) Explain why there is a range of possible values for T which satisfy the requirements of the model. (2)

- (d) Suggest one improvement to the model that would make it more realistic. (1)



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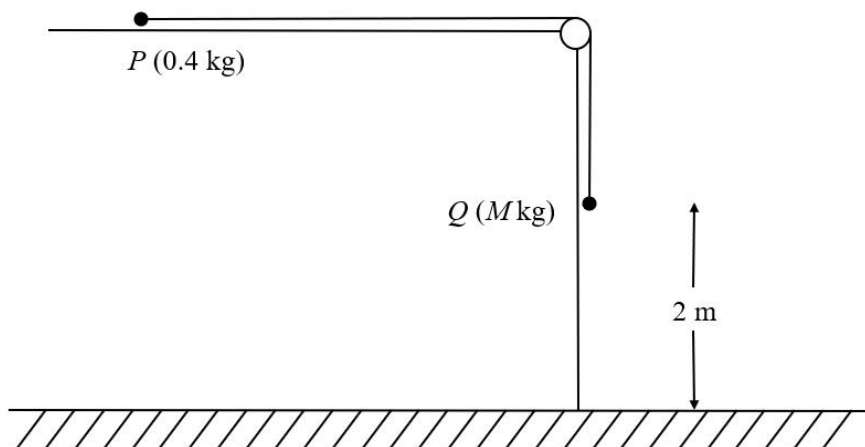


Figure 1

A ball, P , of mass 0.4 kg rests on a rough horizontal table and is attached to one end of a thin rope. The rope passes over a pulley which is fixed at the edge of the table.

The other end of the rope is attached to another ball, Q , of mass $M \text{ kg}$, which hangs freely below the pulley, as shown in Figure 1.

The system is released from rest with the rope taut and with Q at a height of 2 m above the ground and Q moves downwards with acceleration 2.5 m s^{-2}

In the subsequent motion, P does not reach the pulley before Q reaches the ground.

The balls are modelled as particles, the rope as a light and inextensible string and the pulley as being small and smooth.

The total resistance to the motion of P is modelled as having constant magnitude 1.5 N .

The acceleration due to gravity is modelled as being 10 m s^{-2}

Using this model, find, to 2 significant figures,

- (a) (i) the tension in the rope, (6)
 (ii) the value of M , (2)
- (b) the time, after release, for Q to hit the ground. (2)
- (c) State one limitation of the model which will affect the accuracy of your answer to part (a). (1)

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8. A particle P moves along a straight line such that at time t seconds, $t \geq 0$, its velocity, $v \text{ m s}^{-1}$, is given by

$$v = 16 - 3t^2$$

Find

- (a) the distance travelled by P in the first second, (3)
- (b) the value of t at the instant when P changes its direction of motion, (2)
- (c) the value of t at the instant when P returns to its starting point. (3)

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 8 marks)

TOTAL FOR SECTION B IS 30 MARKS
TOTAL FOR PAPER IS 60 MARKS